

CHAITANYA BHARGAVI GONUGUNTLA

Macomb, IL (Open to Relocate) | +1(341) 529-6552 | srichaitanyabhargavi@gmail.com | [Portfolio](#)

PROFESSIONAL SUMMARY

- AI/ML Engineer with 5+ years of experience building production-grade machine learning systems across insurance, healthcare, and financial domains, with strong expertise in supervised learning, deep learning, feature engineering, and model evaluation.
- Proven ability to design and scale end-to-end ML pipelines and real-time systems using PySpark, Kafka, and distributed data processing frameworks to deliver low-latency, high-performance solutions.
- Experienced in model deployment and lifecycle management using MLflow, CI/CD pipelines, Docker, and Kubernetes, ensuring reproducibility, scalability, and consistent production performance.
- Skilled in applying NLP and LLM-based techniques such as Transformer models, BERT, and RAG to develop intelligent systems, achieving up to 18% improvement in model accuracy and 40% reduction in manual effort.

TECHNICAL SKILLS

Programming & Data Processing: Python, SQL, Pandas, NumPy

Machine Learning & Modeling: Supervised & Unsupervised Learning, Deep Learning, Feature Engineering, Model Development & Training, Model Evaluation (Precision, Recall, F1, ROC-AUC), Hyperparameter Optimization, Ensemble Methods, K-Means Clustering, Time Series Forecasting, Bayesian Modeling

ML Frameworks: Scikit-learn, XGBoost, TensorFlow, PyTorch

NLP, Generative AI & LLMs: Transformer Models (BERT), NLP Pipelines (NER, Text Classification), Semantic Search, Embeddings, LLMs, Prompt Engineering, Fine-Tuning (LoRA, PEFT), RAG, LangChain, OpenAI API, Azure OpenAI, Vector Databases (FAISS)

MLOps & Model Lifecycle: MLflow, Experiment Tracking, Model Versioning, Model Monitoring, Drift Detection, CI/CD, Jenkins

Deployment & Backend: FastAPI, Flask, REST APIs, Docker, Kubernetes, Model Deployment

Data Engineering & Processing: Apache Spark, PySpark, Spark Streaming, Kafka, Airflow, ETL Pipelines, Batch & Stream Processing, Distributed Systems

Cloud & Data Platforms: AWS (SageMaker, S3, EC2), GCP (Vertex AI, BigQuery), Azure Machine Learning, Snowflake, Amazon Redshift

Responsible AI & Explainability: SHAP, LIME, Model Interpretability, Bias Mitigation

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Jun 2024 - Mar 2025

Success Tech Solutions Inc | McKinney, TX

- Engineered fraud detection models using Python, Scikit-learn, and XGBoost on insurance claims data, improving prediction accuracy by 18% and reducing false positives in underwriting workflows.
- Leveraged PySpark and Pandas to build feature engineering pipelines that transformed raw claims and policy data into high-quality inputs, improving model generalization and reducing preprocessing time by 30%.
- Assembled scalable ML pipelines using Apache Spark and Kafka to enable batch and near real-time inference, reducing fraud detection latency and improving investigation turnaround time by 25%.
- Developed a document intelligence system using BERT, OpenAI API, LangChain, and FAISS-based RAG, automating claims summarization and classification and reducing manual review effort by 40%.
- Executed hyperparameter tuning using Scikit-learn with experiment tracking workflows, improving model performance and increasing F1-score across validation datasets.
- Delivered low-latency inference services by building REST APIs using FastAPI and Docker, improving response time by 20% and enabling integration with underwriting systems.
- Established reproducible ML workflows using MLflow, CI/CD pipelines, and Kubernetes, improving model version control and reducing deployment cycle time across environments.
- Monitored production model performance using model monitoring frameworks and evaluation metrics such as ROC-AUC and F1-score, identifying data drift early and maintaining consistent prediction accuracy.

Machine Learning Engineer

Jan 2023 - May 2024

Delisis Inc | San Jose, CA

- Devised patient readmission prediction models using TensorFlow and PyTorch on EHR and claims datasets, improving prediction accuracy by 15% and enabling earlier intervention for high-risk patients.
- Mobilized Scikit-learn, XGBoost, and K-Means to build patient segmentation and risk stratification models, improving care prioritization efficiency by 20% across population health programs.
- Structured and maintained ETL pipelines using Apache Airflow, PySpark, and SQL, integrating clinical and operational data sources and reducing data processing time by 35%.
- Engineered real-time anomaly detection workflows using Apache Spark Streaming and Kafka, reducing detection latency by 25% and enabling continuous monitoring of patient vitals.
- Applied NLP pipelines using Transformer Models, BERT, and spaCy to extract structured insights from clinical notes, reducing manual chart review effort by 30%.
- Enhanced model performance through feature engineering and hyperparameter optimization using Scikit-learn with structured experiment tracking, improving F1-score and ensuring consistent validation results.
- Delivered scalable inference services using FastAPI, Docker, and Kubernetes, improving API response time by 20% and enabling reliable model serving across clinical systems.
- Implemented CI/CD pipelines using Jenkins and automated experiment tracking workflows, reducing release cycle time and improving deployment consistency.

Machine Learning Engineer / AI Engineer
Deutsche Borse Group | Eschborn, Germany

Oct 2021 - Sep 2022

- Built fraud detection models using Python, Scikit-learn, and XGBoost on transaction datasets, improving detection accuracy by 16% and strengthening surveillance across trading systems.
- Leveraged PySpark and Kafka to process high-volume streaming data, reducing anomaly detection latency by 30% and enabling faster identification of suspicious activity.
- Designed data integration workflows using Apache Airflow and SQL, consolidating transaction and regulatory datasets and reducing data preparation time by 35%.
- Engineered distributed data processing pipelines using Apache Spark, improving processing efficiency by 40% and enabling near real-time risk analytics.
- Applied NLP pipelines using BERT and spaCy to extract structured risk signals from financial documents, improving compliance review efficiency by 25%.
- Enhanced model performance through feature engineering and hyperparameter optimization using Scikit-learn with experiment tracking, increasing precision and recall across validation datasets.
- Crafted scalable inference services using Flask and Docker, improving response time by 20% and enabling integration with internal risk management systems.
- Implemented model monitoring workflows using MLflow and evaluation metrics such as ROC-AUC and F1-score, maintaining production performance and reducing model drift impact.

Data Scientist

Jun 2017 - Sep 2019

Hetero Healthcare Limited | Chennai, India

- Analyzed healthcare supply chain and sales data using Python, SQL, Pandas, and NumPy, uncovering demand patterns that improved inventory planning accuracy by 18% across regional distribution centers.
- Built demand forecasting models using Scikit-learn and XGBoost, reducing stockouts by 22% and improving product availability across multiple warehouse locations.
- Designed data transformation workflows using Pandas and SQL, improving data quality and reducing inconsistencies by 30%, enabling reliable downstream analytics and reporting.
- Developed integrated analytical datasets using SQL and Python, reducing data preparation time by 25% and enabling faster access to insights for business teams.
- Created interactive dashboards using Power BI, improving visibility into sales and inventory KPIs and accelerating decision-making for supply chain operations.
- Automated reporting pipelines using Apache Airflow and SQL, reducing manual effort by 35% and ensuring consistent and timely data refresh cycles.
- Performed statistical analysis using Python and NumPy, identifying key drivers of demand variability and improving the accuracy of forecasting inputs.
- Collaborated with business stakeholders to build analytical solutions using Python and Scikit-learn, enabling data-driven decisions and supporting early adoption of predictive analytics.

PROJECTS

End-to-End Fraud Detection System with Real-Time Inference

- Built a scalable fraud detection pipeline using PySpark, Kafka, and XGBoost, processing high-volume transaction streams and improving detection accuracy by 17% while reducing false positives.
- Designed feature engineering and model optimization workflows using Pandas, Scikit-learn, and MLflow, improving F1-score and ensuring stable model performance across validation datasets.
- Deployed real-time inference services using FastAPI, Docker, and Kubernetes, enabling low-latency predictions and ensuring high availability across distributed environments.

Healthcare Risk Prediction & Explainable AI System

- Developed patient readmission prediction models using TensorFlow, PyTorch, and Scikit-learn, improving prediction accuracy by 14% across clinical datasets.
- Applied explainability techniques using SHAP and LIME to interpret model outputs, improving transparency and supporting adoption by healthcare stakeholders.
- Integrated models into production using MLflow, CI/CD pipelines, and Docker, ensuring reproducible deployments and consistent model performance.

RAG-Based Intelligent Document Processing System

- Designed a document retrieval and generation pipeline using LangChain, OpenAI API, and FAISS, enabling context-aware responses and improving retrieval accuracy.
- Implemented semantic search using embedding models and vector databases, enhancing document relevance and reducing search latency across large datasets.
- Improved response quality using prompt engineering and fine-tuning techniques (LoRA, PEFT), increasing output consistency and contextual accuracy.

EDUCATION

Master of Science in Data Science
University of Trier | Germany

Oct 2019 - Jun 2022

CERTIFICATIONS

- Machine Learning Specialization - **Coursera**
- Deep Learning Specialization - **Coursera**
- Generative AI with Large Language Models - **Coursera**
- MLOps Specialization - **Coursera**
- AI Engineering Foundations: Machine Learning - **LinkedIn Learning**
- Building Generative AI Applications with LLMs - **LinkedIn Learning**